$A=[0;2]$
$B=[4;5]$
$C=[2;-7]$

a) A, B, C jsou vrcholy $\triangle$?

$\overrightarrow{AB}=(4;3)$
$\overrightarrow{AC}=(2;-9)$

$$\left.\begin{aligned} 4&=2k \Rightarrow k=2 \\ 3&=-9k \Rightarrow k=-\tfrac{1}{3} \end{aligned}\right\} 2 \neq -\tfrac{1}{3} \Rightarrow A, B, C \text{ jsou vrcholy } \triangle$$

b) $X=[\ ;\ ]$

$\overrightarrow{BX}=3\overrightarrow{AC}-2\overrightarrow{BC}$

$X-B=3(2;-9)-2(-2;-12)$
$X=3(2;-9)-2(-2;-12)+[4;5]$
$X=(6;-27)+(4;24)+[4;5]$
$X=[14;2]$

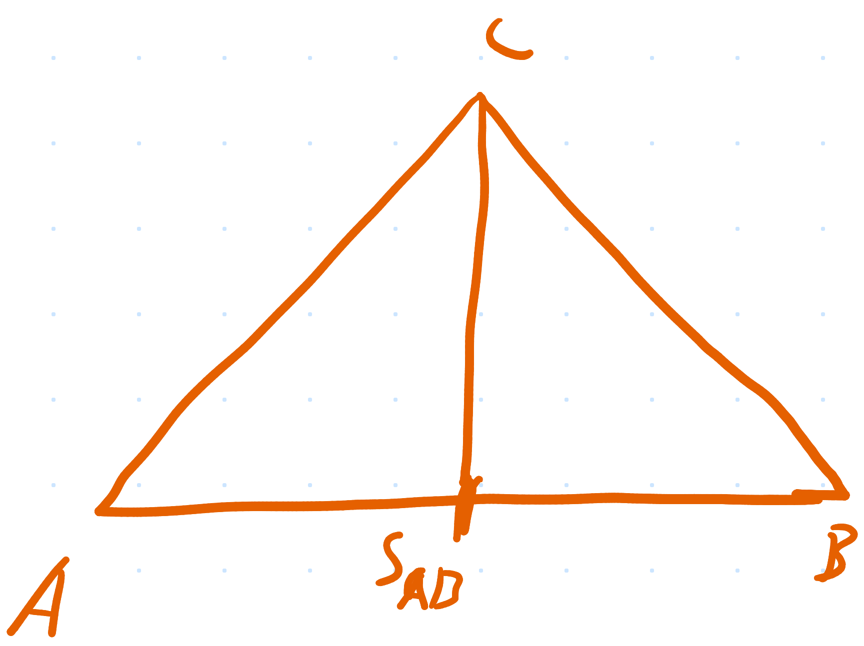
$A=[0;-8]$
$B=[1;2]$
$C=[-4;4]$

a) A, B, C vrcholy $\triangle$?

$\overrightarrow{AB}=(1;10)$
$\overrightarrow{AC}=(-4;12)$

$$\left.\begin{aligned} 1&=-4k \Rightarrow k=-\tfrac{1}{4} \\ 10&=12k \Rightarrow k=\tfrac{5}{6} \end{aligned}\right\} -\tfrac{1}{4} \neq \tfrac{5}{6} \Rightarrow \text{jsou vrcholy}$$

b) velikost t_c



$S_{AB}=[\tfrac{1}{2};-3]$

$|t_c|=|CS|=\sqrt{20{,}25+49}=\sqrt{69{,}25}$

$A=[0;-8]$
$B=[1;2]$
$C=[-4,4]$

c) bod D, aby ABCD byl rovnoběžník

$D = C + \overrightarrow{AB}$

$= [-4;4] - (1;10)$

$= [-5;-6]$

d) ověřte zda $\overrightarrow{AB} \perp \overrightarrow{BC}$ $(u_x v_x + u_y v_y = 0)$

$\overrightarrow{AB} = (1;10)$
$\overrightarrow{BC} = (-5;2)$

$1\cdot(-5) + 10\cdot 2 = 0$
$-5 + 20 = 0$
$15 \neq 0$

nejsou kolmé

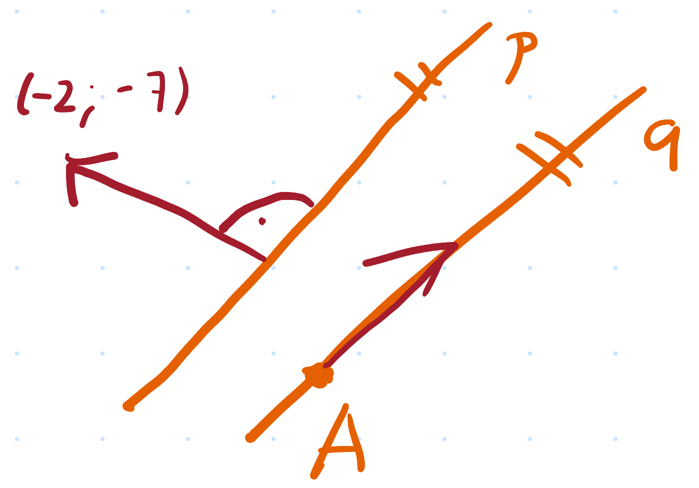
a) k dané přímce p sestavte parametrickou rci rovnoběžky q.

$A=[-3;-4] \in q$
$p: -2x-7y+4=0$

$\vec{p} = (-2;-7) \perp (7;-2)$

$x = a_x + u_x \cdot t$
$y = a_y + u_y \cdot t$

$q: x = -3 + 7t$
$y = -4 - 2t$



b) k přímce p napište obecnou rci kolmice q.

$A=[0;0]\in q$

$p: x = 1+3t$

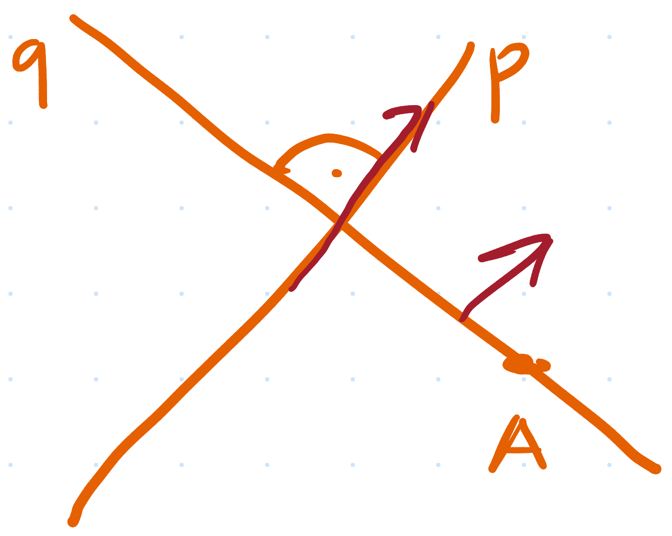
$y=-5+8t$

$ax+by+c=0$

$p=(3;8)$

$3x+8y+c=0$

$c=0$



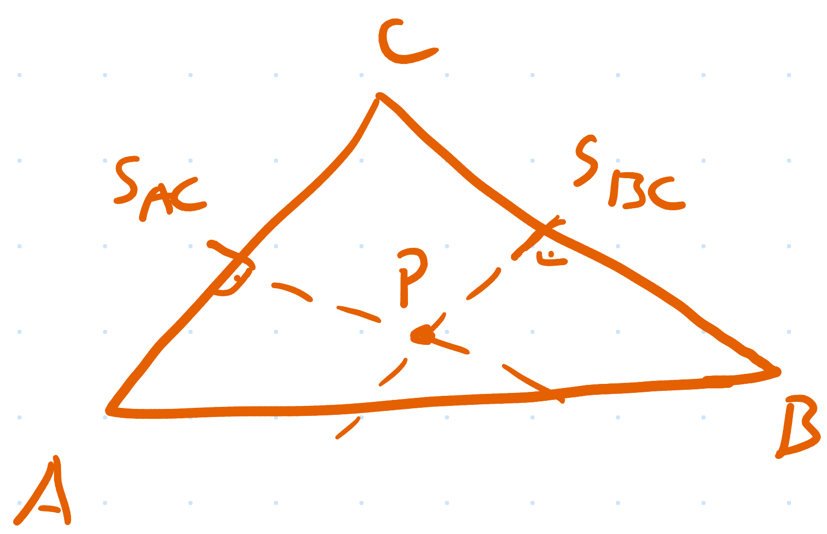
$q: 3x+8y=0$

a) body A, B, C jsou vrcholy Δ. Vypočtěte souřadnice průsečíku os stran AC, BC

$A=[9;-8]$

$B=[3;-2]$

$C=[5;-5]$



$S_{AC}=[7;\frac{13}{2}]$

$\vec{AC}=(-4;3)$

$-4x+3y+c=0$

$-28+19{,}5+c=0$

$c=8{,}5$

$-4x+3y+8{,}5=0$

$S_{BC}=[4;-3{,}5]$

$\vec{BC}=(2;-3)$

$2x-3y+c=0$

$8+10{,}5+c=0$

$c=-18{,}5$

$2x-3y-18{,}5=0$

$20+3y+8{,}5=0$

$3y=-28{,}5$

$y=-9{,}5$

$-4x+3y+8{,}5=0$

$2x-3y-18{,}5=0$

$-2x \quad -10=0$

$-2x=10$

$x=-5$

$P=[-5;-9{,}5]$

b) je daná přímka p a bod AB vypočítejte souřadnice středu kružnice, který leží na p a kružnice prochází A, B.

$p: x = 3t$
$\underline{y = 7 + 7t}$

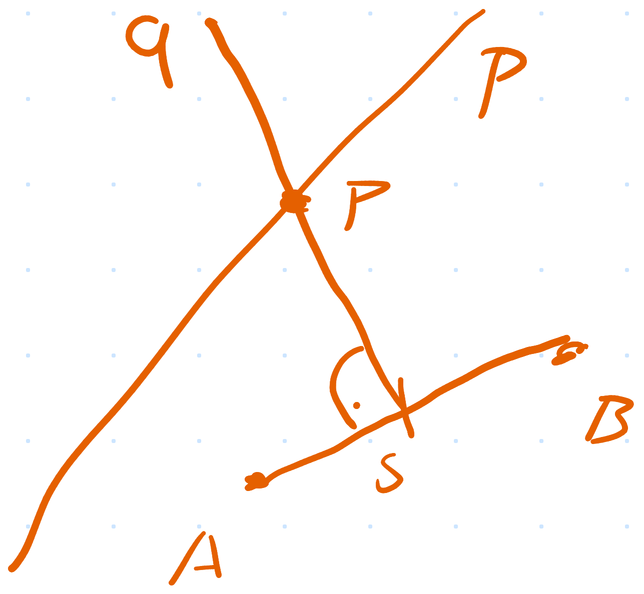
$A = [3; -3]$
$B = [8; 0]$

$S_{AB} = [5{,}5; -1{,}5]$
$\overrightarrow{AB} = (5; 3)$

$5x - 3y + c = 0$
$5(5{,}5) - 3(-1{,}5) + c = 0$
$27{,}5 + 4{,}5 + c = 0$
$c = -32$

$5x - 3y - 32 = 0$

$5 \cdot 3t - 3(7 + 7t) - 32 = 0$
$15t - 21 - 21t - 32 = 0$
$-6t - 53 = 0$
$-6t = 53$
$t = -\frac{53}{6}$



$x = 3t = 3 \cdot \left(-\frac{53}{6}\right) = -26{,}5$
$y = 7 + 7\left(-\frac{53}{6}\right) = 7 + 61{,}83$
$= 68{,}83$

$P = [-26{,}5; 68{,}83]$